



Figure 28-10 Shifting an animal's center of gravity by lifting a limb off the ground and forcing the patient to redistribute its weight.



Figure 28-12 Dynamic weight shifting may also be performed while the animal is walked in circles.



Figure 28-11 Dynamic balance and proprioception may be challenged while gently shifting the dog's weight while walking. Caution should be used to avoid too much force so the patient does not fall.

but instead will bear the weight on the handler's hand or collapse to the ground. A technique to avoid bearing all of the weight on the handler is to slowly abduct the raised limb, which allows transfer of weight from the handler to the desired limb to avoid falling.

Weight shifts may also be performed during walking to provide challenges to dynamic stability. As the animal is walked in a straight line, the handler gently bumps or pushes the animal to one side to challenge the dog to maintain its balance (Figure 28-11). Caution should be used to tailor the force of the push to the animal's stage of recovery to avoid falls and injury. As the patient improves, dynamic weight shifting may be performed while the animal is walked in circles, figure eights, or while pole weaving (Figure 28-12).

After the animal has reasonable balance and proprioception while standing and walking, more challenging balance and proprioception exercises may be initiated.

Balance Board

A board placed over a fulcrum, such as a length of pipe, may be used to rock the dog forward and backward or side to side (Figure 28-13). Movement of the surface the dog is standing on provides additional challenge to the patient. In most cases, one person helps to support the dog while standing on the board, while another person slowly rocks the board back and forth. Initially, the movements are slow and rhythmic. As the patient improves, the motions may be faster and less predictable to challenge the animal to react quicker.

Wobble Board or Quadruped Balance Platform

A wobble board or human biomechanical ankle platform system board may be used to help the animal practice proprioceptive training on just the forelimbs or the hindlimbs (Figure 28-14). If the goal is to challenge the animal using all four limbs, a balance board large enough to accommodate all four limbs may be used (Figure 28-15). It is important to provide support to allow the animal to shift its weight and exercise its proprioception and balance. Again, the initial movements are slow and rhythmic, but as the patient improves, the motions may be faster and less predictable to further challenge the animal.

Exercise Balls and Rolls

Therapeutic exercise balls and rolls designed for human use may be employed to improve balance, coordination, and strength. The forelimbs may be placed on the ball and the patient supported by the handler, requiring the dog to maintain static balance of the caudal trunk and rear limbs

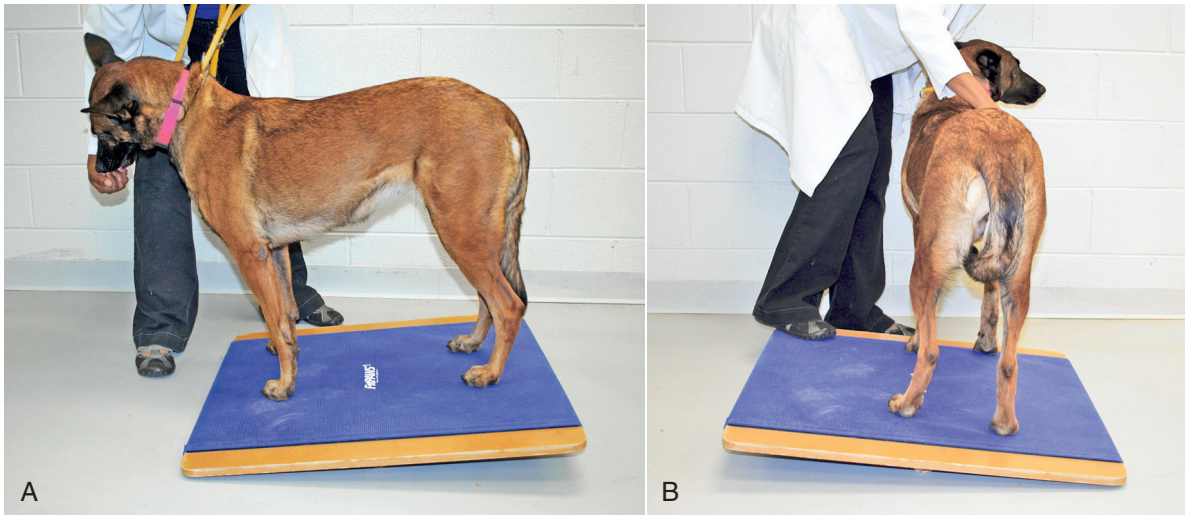


Figure 28-13 A balance board may be used to provide balance and proprioceptive training from the forelimbs to the hindlimbs (A) or side to side (B).



Figure 28-14 Use of a wobble board or human biomechanical ankle platform system board under a patient's hindlimbs to assist with balance and proprioceptive training. Similar training may be applied to just the forelimbs, or small dogs may stand with all four limbs on the board.

(Figure 28-16). Dynamic balance may also be challenged as the ball or roll is slowly moved forward, backward, and side to side, challenging the rear legs to maintain balance while movement occurs. To address the cranial trunk and forelimbs, the rear limbs are placed over the ball as the forelimbs are asked to balance the body weight during both stance and gentle movements (Figure 28-17).

Further challenges may be instituted by having the dog stand on a therapy ball or roll (Figure 28-18). This is extremely challenging and uses multiple core stabilizing muscles. Therefore the sessions should be relatively short to avoid fatigue and possible injury. One person stabilizes the ball, while one or two people, depending on the size of the dog, support the patient. Further challenges are



Figure 28-15 Using a balance platform to help quadruped animals develop improved balance and proprioception.

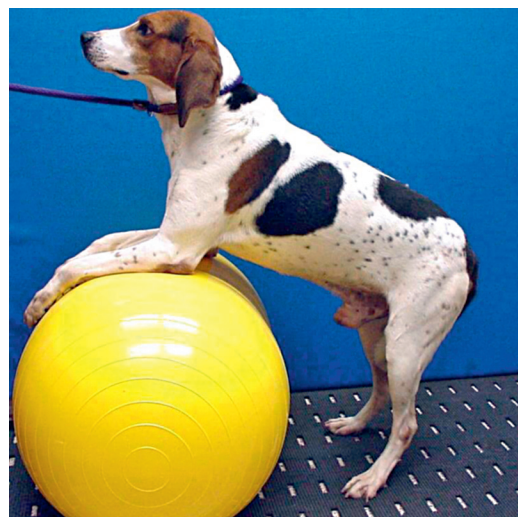


Figure 28-16 An exercise roll may be used to challenge balance and proprioceptive function. The forelimbs may be placed on the ball and the patient supported by the handler, requiring the dog to maintain static balance of the caudal trunk and rear limbs.



Figure 28-17 To address the cranial trunk and forelimbs, the rear limbs are placed over the exercise roll as the forelimbs are asked to balance the body weight during both stance and gentle movements.



Figure 28-18 A very challenging balance and proprioceptive exercise is to have the animal balance on an exercise roll with all four limbs. Support must be provided by the therapist to avoid possible injury.

provided by slowly rocking the therapy ball back and forth or side to side.

Functional Training Exercises for Balance and Proprioception

Standing or walking on foam rubber, mattresses, air mattresses, or trampolines alters the texture of the ground and challenges the animal's functional walking proprioceptive ability (Figure 28-19). Animals may also be walked over different surfaces, such as grass, concrete, a mulched path, a curb, sand, or other surfaces.



Figure 28-19 Standing or walking on foam rubber, mattresses, air mattresses, or trampolines alters the texture of the ground and challenges the animal's functional balance and proprioceptive ability.



Figure 28-20 Walking through vertical weave poles helps develop coordination and balance while learning to turn and pivot.

Animals must also develop balance and proprioception while performing activities of daily living, including turning, walking over obstacles, and negotiating stairs. Initially, dogs may be walked in large circles, going in both directions. The radius of the circle may be reduced as the animal improves, and patients may then progress to walking in a figure-eight pattern. Animals may also benefit by walking through vertical weave poles (Figure 28-20). In addition to helping improve general proprioception, patients also begin to learn how to contract their limb muscles to help stabilize joints while slowly pivoting and turning.

Some dogs must also perform certain activities in their home environments, such as negotiating stairs or walking over obstacles. Therefore walking up and down stairs may be an important activity to challenge proprioception in some patients. If possible, dogs may begin training on stairs with a gradual rise, walking slowly, and then progress to steeper stairs. Dogs may also walk over cavaletti rails to begin to judge how high to lift their legs to step over an obstacle. The height of the rails may vary between different strides, or the rails may be crossed at different



Figure 28-21 Walking over cavalletti rails helps dogs learn how to negotiate obstacles and walk over them by lifting their limbs to the appropriate height.



Figure 28-22 Walking over cavalletti rails while negotiating vertical weave poles provides additional challenges to balance and proprioception.

angles to provide additional challenges (Figure 28-21). Finally, dogs may negotiate vertical weave poles while stepping over cavaletti rails to provide additional challenges (Figure 28-22).

Summary

A rehabilitation program for a patient with proprioceptive and balance deficits does not need to be elaborate or costly. Consideration should be given to patient, owner, and therapist needs. Although the willingness of the owners to help with rehabilitation and the education level of the staff influence the rehabilitation program the most, some rehabilitation is better than none at all.

Therapeutic exercises are undoubtedly the most important aspects of rehabilitation. They may (and should) be incorporated with other modalities to enhance recovery, but using exercises that help improve proprioception and balance are among the earliest aspects of canine rehabilitation. In addition, most of these exercises help improve muscle strength, joint mobility, and limb use. Although a variety of techniques have been described, the ingenuity of the rehabilitation team, including the owner, will allow the development of other exercises that are specific to a patient's recovery. The keys to a successful therapeutic exercise program are to have site- and condition-specific exercises whenever possible, to use a variety of exercises and techniques to keep the therapy team and patient from becoming bored, and to allow the animal to appropriately progress to the next phases of rehabilitation.

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29 Therapeutic Exercises: Early Limb Use Exercises

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Therapeutic exercise is perhaps the most valuable modalities used in canine physical rehabilitation. Therapeutic exercise is an important method to assist an animal's return to the best function possible. In addition, the equipment needed is relatively inexpensive, and similar principles apply to a variety of individuals and conditions. Therapeutic exercise programs designed for the home environment also provide an opportunity for owners to become actively involved in their pet's rehabilitation.

Some of the common goals of early limb use exercises are to improve active pain-free range of motion; muscle mass and muscle strength to allow weight bearing, balance, and performance of activities of daily living; lameness; aerobic capacity; and weight loss and to help prevent further injury.¹⁻⁴

When designing a therapeutic exercise program, several factors must be considered. A problem list is developed based on an initial evaluation of the rehabilitation patient, realistic outcome goals are established, and a treatment plan is formulated to address the identified problems. Therapeutic exercise is a significant component of the treatment plan. Appropriate exercises are those that can be performed safely and effectively and accomplish the therapeutic goals. When prescribing therapeutic exercise, the therapist should understand the diagnosis, identify the structure or structures involved, and recognize the stage of tissue recovery with the resultant functional limitations. It is important to remember that the force placed on the body with exercise should not exceed the strength of healing tissues.

The focus on patients that have serious afflictions with limited to no weight bearing on the affected limbs centers on early limb use and is the topic of this chapter. As patients are able to bear weight and begin to use their limbs, the emphasis shifts to improving proprioception and balance for patients with neurologic conditions or those recovering from joint injury or surgery, as described in Chapter 28. Finally, as patients improve with an approach to maximal expected function, the emphasis should be on strength, speed, and joint motion exercises, which are the focus of Chapter 30.

Treatment considerations and the choice of exercises vary with each stage of tissue repair and endurance. As the

animal improves clinically and tissue healing progresses, the exercise plan should be altered to match the animal's progress and appropriately challenge the involved tissues. The intensity of an exercise may be increased or reduced by changing the duration of time that an animal performs an exercise, the frequency with which an exercise is performed, and the rate or speed at which a particular exercise is performed (Box 29-1). Any of these may be altered to fine-tune an exercise prescription to achieve the expected outcome goals. In general, the frequency of the activity is increased first if the patient is severely affected. For example, a realistic initial goal for a morbidly obese, deconditioned patient with degenerative joint disease may be to increase the amount of time the dog is able to comfortably tolerate walking by increasing the frequency of leash walks from twice daily to three times daily and finally to four times daily. This strategy helps to improve endurance and promote weight reduction, while limiting fatigue or muscle soreness. Increasing the length or duration of activity is usually the next phase of activity. For example, the dog may progress from walking 5 minutes four times per day to 6 minutes four times per day the next week. Increasing the speed at which the dog walks may not be a realistic initial goal for this animal at this point. A contrasting example may be an athletic animal recovering from injury that must be challenged to improve speed and endurance to meet the goal to return to performance activity. This type of patient may progress from jogging 20 minutes per session to playing ball in an enclosed run. It is important for the therapist to have an understanding of exercise intensity and what is appropriate for each patient during rehabilitation treatment. Routine reevaluation of the patient is recommended to assess the adaptations that are occurring with the rehabilitation treatment plan and to determine the appropriate rate of progression.

Common activities to encourage early limb use include standing exercises, controlled leash activities, ground treadmill activity, taping a bottle or syringe cap to the bottom of an unaffected foot to encourage weight bearing, slinging the contralateral good limb, and using balance balls or rolls. The exercise plan should target the affected muscles and other structures. The type of injury or repair must be considered when formulating a treatment plan. For example, the postoperative treatment of hip dysplasia

Box 29-1 Therapeutic Exercise Strategies

Exercise is generally increased as the patient and healing tissues allow:

1. Increase frequency of activity (e.g., from twice daily to three times daily, and then to four times daily).
2. Increase the duration of activity (e.g., from 5 minutes to 6 minutes).
3. Increase the speed or intensity of activity (e.g., increasing from a walk to a slow jog, and then from a slow jog to a medium trot).

In general, it is safe to increase the amount of activity 10-15% per week, as long as there is no increase in pain, decreases in function, etc.

differs depending on whether a femoral head and neck excision or a total hip replacement was performed. With this knowledge and understanding, appropriate decisions can be made regarding therapeutic exercise choices.

Therapeutic exercise routines should be monitored at regular intervals by a trained individual familiar with the patient and the exercise techniques. Inappropriate exercise or improper technique may result in inappropriate stresses, further injury, or exacerbation of an existing condition. Certain exercises may not be safe for the strength, flexibility, or endurance level of the animal performing the exercise, or an exercise may not be the correct one to accomplish the intended goal.

Standing Exercises**Assisted Standing Exercises**

Patients with severe injuries or debilitating conditions may not be able to stand and support their own body weight. A period of standing, either completely or partially assisted, may strengthen the patient, aid in proprioceptive training, improve circulation and respiration, provide an opportunity to eliminate, and enhance the patient's psychological well-being. The purposes of assisted standing exercises are to encourage neuromuscular function, reeducate muscles, develop strength and stamina of supporting postural muscles, and enhance proprioception.

Maximal assisted standing is appropriate for patients that are unable to support their body weight as a result of paralysis, paresis, pain, trauma, postoperative precautions, or general debilitation. Maximal assistance may be defined as support of 75% to 100% of the patient's body weight by the handler to maintain a standing posture. Animals with multiple orthopedic injuries (especially pelvic fractures following repair or after a period of healing), neurologic conditions, or severe debilitation are excellent candidates for assisted standing exercises (Figure 29-1). Assisted standing exercises are among the first therapeutic exercises prescribed for severely afflicted animals that are



Figure 29-1 Support while standing is beneficial for patients with limited ability to stand and bear body weight. A sling may be used to provide support for maximal assisted standing of a patient.

unable to independently rise from a recumbent position or support their own body weight, and may begin in patients that are receiving adequate pain control following injuries that are stable. Animals with unstable injuries or in pain resist standing and may further injure themselves or the therapist while struggling.

It is appropriate to begin assisted standing exercises in patients with adequate muscle tone and some ability of the limbs to initiate motion and resist collapse of the body (adequate upper motor neuron function). Patients with lower motor neuron signs have little or no muscle tone and flaccidity of the limbs. Although patients with lower motor neuron conditions will likely benefit from standing exercises, it is unlikely that they will be able to fully support their weight, and some additional assistance will be required.

Depending on animal size and weight, the handler can manually lift or use an assistive device such as a pelvic sling, abdominal sling, or thoracic sling to assist the animal from a recumbent to a supported standing position. To perform standing exercises, a sling is placed under the cranial thorax, caudal abdomen, or both for conditions affecting the forelimbs, rear limbs, or both, respectively. After the animal is in a supported standing position, the handler or a second person manually places the animal's limbs in a normal stance position, with the feet positioned squarely on the ground. Even distribution of the animal's weight on the ground allows for greater stability and may provide position sense, which provides proprioceptive input and facilitates the animal's awareness of its joints.

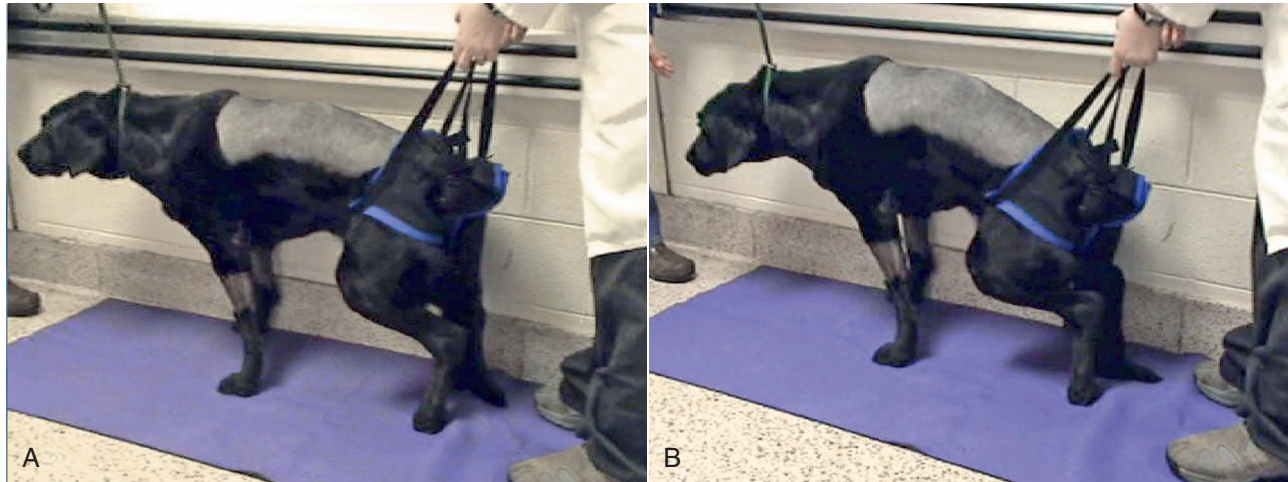


Figure 29-2 Tetraparesis or other conditions causing weakness in multiple limbs. **A**, Use of a sling to help provide support for standing. **B**, As the patient begins to fatigue and attempts to sit, the therapist lifts the dog back into a standing position.

Additional limb awareness can be encouraged via proprioceptive neurosensory facilitation techniques such as tickling the paw or distal segments, ascending circular touching, or back brushing the hair of the limbs.

The animal should be encouraged to support its body weight as it is able, and the therapist gives only the necessary assistance to maintain the standing position (Figure 29-2, A). While supporting the animal, the handler slowly releases tension on the assistive device, allowing the animal to continue weight bearing as much as it is able. Alternatively, it may be easier for the therapist to support the patient manually with a hand placed on the ventral pelvis to maintain the rear limbs in a normal position, preventing inappropriate crossing or a base narrow stance. Because the animal is encouraged to accept and support a portion of its weight, strength, balance, coordination, and proprioception are challenged. If the dog begins to collapse, the sling is gently pulled up to assist the patient back into the standing position and the exercise is repeated (Figure 29-2, B). It is important to place the limbs in a square standing position to give neuromuscular feedback and to also allow the animal to stand in a stable fashion.

During a treatment session, the animal may need periods of rest. Although the duration of each session should match the animal's tolerance, an initial starting goal is 5 to 15 repetitions two to three times daily, gradually increasing to standing 5 minutes per session. Individual patients progress at different rates. The therapist can keep track of and document the amount of time the animal is able to stand and how much assistance is needed.

Active Assisted Standing

As the animal becomes stronger and regains neuromuscular function, maximal assistance from the handler is not required. The progression from maximal assisted standing

to active assisted standing occurs when the animal is able to actively support some of its weight, requiring only partial assistance from the handler (less than 75% of body weight). The animal should be encouraged to support as much of its weight as possible. The handler provides only the additional assistance required to maintain a standing position or prevent loss of balance. The amount of assistance required may be documented by recording the actual weight the handler is required to support for the animal to maintain a standing position, as well as the length of time the animal can maintain a standing position. The animal may require periodic rests during a treatment session. As the animal's strength and endurance improve, the duration of each session may be increased, the number of rest periods decreased, and the amount of assistance decreased.

Standby Assisted Standing

The next progression in standing activities is standby assisted standing. At this point of the recovery, the animal has the strength and motor control necessary to support itself against gravity in a standing position. However, it may still experience ataxia or weakness and have an occasional loss of balance, requiring standby assistance. The therapist should be behind the animal or at the animal's side, ready to guard against a fall caused by a loss of balance (Figure 29-3). The therapist does not assist the dog unless support is needed to prevent a fall. As strength improves, the patient will be more willing to advance the limbs if motor function is present. In cases of hindlimb paresis, the therapist may also challenge the patient further by lifting a forelimb to encourage weight bearing on the hindlimbs. It is also beneficial to lift a forelimb in dogs with unilateral hindlimb weakness to shift the weight to the hindlimbs.



Figure 29-3 The therapist is ready to provide support to a patient in the event of a loss of balance. The therapist does not assist the dog unless support is needed to prevent a fall.

Aquatic Standing

Standing in water may be a valuable activity for large patients with generalized weakness, lower motor neuron disease, or other conditions resulting in inability to stand and support weight. The buoyancy of the water often results in a patient's willingness to try to stand and bear some weight; often these patients will not even try to stand on dry ground. Supportive devices, such as life vests, slings, or harnesses may be used to provide further assistance and support. The reader is referred to Chapter 31 on aquatic therapy for further details.

Aids for Assisted Standing

Body Slings

Nonambulatory patients may benefit from placement in a body sling or harness to provide support for standing (Figure 29-4). It may be easier to use a lift device in conjunction with a sling to help support larger patients (Figure 29-5). An advantage of a sling is that the patient is able to maintain a body position with the limbs placed under the body in a standing position with partial or complete support of body weight, which provides an opportunity for limb strengthening and early proprioceptive training. Frequent episodes of supported standing also help to relieve pressure on bony prominences and may reduce the chances of decubital ulcer formation and pulmonary complications. Excessive congestion of the lungs that may result from spending long periods in lateral recumbency may be prevented with early implementation of standing in a sling. The body sling must be adjusted properly so that respiration is not compromised and the limbs are not compressed in the openings of the sling. Proper padding is also important for comfort. The skin should be inspected after each session in the sling to identify areas of potential skin irritation or breakdown.



Figure 29-4 A sling that may be used to assist standing. Standing helps the patient develop stamina and endurance with minimal assistance required from a therapist.



Figure 29-5 A lift device may be used in conjunction with a sling to help support larger patients.

Rubber bungee cords may be used to secure the sling to the supporting frame. This may aid in the facilitation of weight bearing and movement during the sling sessions. The elasticity of the cord permits some gentle up-and-down movement that encourages the animal to safely bear partial weight on its limbs. It also may allow some attempts to move forward, backward, and side to side. A gentle bouncing motion may be manually introduced by the therapist to initiate rhythmic stabilization of muscle groups throughout the trunk and limbs.



Figure 29-6 A variety of exercise rolls exist so that the appropriate size may be picked for nearly any size patient.



Figure 29-7 It is sometimes helpful to have very small dogs straddle a foam roll or roll of paper towels. In this photo, a dog is straddling a foam roll to facilitate neuromuscular electrical stimulation.

Exercise Rolls

A properly fitted inflatable exercise roll (physioroll, peanut) or a bolster may also be used to actively assist the animal with weight bearing, balance, coordination, strength, and weight shifting. The advantage of this form of assisted standing is that patients may stand independently, stand for longer periods, and avoid the problems inherent with long-term standing in a sling, such as compression of the upper limbs as they exit through the holes of a sling. A variety of exercise rolls exist so that the appropriate size may be picked for nearly any size patient (Figure 29-6). In very small dogs, it is sometimes helpful to have them straddle a foam roll or roll of paper towels (Figure 29-7). Using the roll as an assistive device also relieves stress on the handlers. Unless the animal is small, this technique may require two people to properly assist the patient into position.

Patients with limited ability to bear their own weight may be placed over an exercise roll such that the body straddles the roll with weight supported by the roll. The animal is placed in a standing position over an appropriately sized exercise roll (the roll provides greater lateral support and stability than a round exercise ball). The size



Figure 29-8 Patients with limited ability to bear their own weight may be placed over an exercise roll such that the body straddles the roll with weight supported by the roll. The animal is placed in a standing position over an appropriately sized exercise roll that may be used to provide additional challenges or increase weight bearing on the front or rear limbs or to assist standing (the roll provides greater lateral support and stability than a round exercise ball). The size should allow the animal to touch the ground with all four feet.

should allow the animal to touch the ground with all four feet (Figure 29-8). If the roll is too tall, it may be deflated to meet the animal's height. Rolls that are slightly deflated are softer, more stable, and easier to work with than those that are fully inflated and very firm. The softer roll conforms to the animal's body better and is more comfortable. After the animal is secure on the roll, one person stabilizes the front of the animal and another, if needed, stabilizes the rear.

While the animal is supported in a standing position with paws on the ground, the handler can generate a very gentle up-and-down bouncing motion through the patient and the inflated roll. This provides proprioceptive input and may stimulate contraction of the supporting limb muscles. Dynamic balance and weight bearing may also be challenged in the thoracic or pelvic limbs as the therapists shift the roll forward and backward. As the animal becomes stronger, these same techniques may be performed at faster speeds to provide greater challenges to neuromuscular function and balance. It is recommended to perform standing over a roll during mealtimes, to allow the patient to focus on food and not on the therapist. This also provides positive reinforcement and relaxation for anxious patients and tends to decrease the extensor rigidity for brief periods in patients with upper motor neuron signs.

Carts

Mobility carts and their associated slings may be used as assistive devices for active assisted activities. Several manufacturers offer supportive carts for dogs, cats, and other animals. These carts are designed to assist with ambulation



Figure 29-9 A four-wheeled cart may be used for assisted ambulation in patients with tetraparesis.

of animals that are unable to support adequate weight for ambulation. Common indications for cart use are paralysis, paresis, severe arthritis, trauma in multiple limbs, and generalized weakness affecting one or more limbs. Carts allow the animal to have some degree of independence and reduce the lifting and support needed from a caregiver. Two-wheeled carts may be used for patients with active forelimb function, and four-wheeled carts are appropriate for patients with all limbs affected (Figure 29-9).

It is important that the cart be properly sized and fitted to ensure comfort and adequate function. Most manufacturers provide specific instructions regarding proper measurements to provide a custom fit. The therapist can assist with the measuring and fitting process. In addition, the cart should have adequate padding for comfort and to prevent chafing or breakdown of skin, but it should not interfere with the use of functional limbs.

If the animal has some functional use of the limbs, the devices should be adjusted to the patient's height so the feet may be placed (either manually by the therapist or independently by the animal) in a normal weight-bearing position on the ground. The patient should not dangle above the ground in the cart because this may compromise circulation and place undue pressure on soft tissues that are in contact with the supporting structures of the cart. Patients with poor motor or proprioceptive function may benefit from foot protection (rubber booties or infant socks) to prevent trauma to the dorsum of the paws. Patients with no motor function will likely need to have their feet maintained in a non-weight-bearing position.

Dynamic Ambulation Activities

Assisted Ambulation and Gait Training

If a dog is unable to walk independently, an assistive device may be used to support the animal as needed. Gait training



Figure 29-10 Commercial canine treadmills are available for rehabilitation.

should begin with the use of a fabric or elastic band sling, harness, or canine cart. The handler should assess the animal's needs before choosing the appropriate device. Encourage the dog to move slowly, allowing time for the dog to advance the limbs as independently as possible. Allow adequate time for the feet to come in contact with the ground with each step during the stance phase of gait. It may be necessary for the handler to manually assist the dog in the sequencing and placement of the limbs as the animal relearns the walking gait. If the animal moves too quickly, it will often avoid bearing weight on a painful limb, adopting an abnormal gait, such as a three-legged lameness, hopping, or dragging of a limb. The emphasis is on weight bearing with each and every step, by walking slowly. It is important for each person handling the dog to be consistent and not allow the dog to move too quickly. For food-motivated dogs, choosing a highly desirable treat that is only used when encouraging movement can help positively associate movement or the active attempts at movement during a therapy session. In some cases, dogs may be stimulated to ambulate to a greater degree if other animals are present and a game of "assisted chasing" ensues.

Treadmill Walking

Walking on a treadmill is very useful in rehabilitation. Most dogs trained to a leash readily take to treadmill walking in one or two sessions. A variety of treadmills are available for use, including commercial canine treadmills (Figure 29-10). A number of models available for human use may be modified for canine use by adding an overhead bar with a support system to which a canine harness can attach. A harness is useful to help support the dog in case it stumbles or falls. Alternatively, a therapist may support the dog with a sling while on the treadmill. Side rails or fences placed on both sides of the treadmill are useful if a dog tends to step off to the side. Other useful features include variable speed control, a timer, and the ability to change the incline or decline of the surface.

Treadmills are useful for patterning gait and encouraging initial weight bearing following surgery. When the dog stands with the foot carried near the ground, it will generally begin to bear weight when it is walked on a treadmill. It is important that the treadmill not face toward a wall; rather, it should face toward a hallway or the middle of a room to encourage unimpeded walking. Having one person standing in front of the dog with words of encouragement or treats and one person straddling the dog behind is helpful in the early training stages to keep the dog walking straight.

Walking on a treadmill for the first time is a new and sometimes awkward experience. Animals are not used to the ground moving under them, so when they walk on a moving belt, proprioception, coordination, and balance are challenged. The ground moving underneath the dog often encourages a dog that is non-weight bearing on a limb as a result of an orthopedic condition to begin using the limb. In many instances, the animal will continue to use the limb even after the treadmill session is over. For patients with neurologic conditions, it may be necessary to support the dog with a sling or harness. The therapist may stand or squat beside a dog and manually advance a foot during the normal gait sequence to correct abnormal foot placement, such as knuckling over, and encourage gait reeducation.

Assisted active advancement of a limb during the swing phase of gait can also be performed by using an elastic band. An animal with conscious proprioceptive deficits or muscle weakness may have difficulty advancing the limb during the swing phase of gait; these patients can be assisted by using an elastic band to help advance the limb. After one end of the elastic band is attached to the foot or just above the hock, the other end is secured to the body with the direction of pull designed to assist active advancement of the limb during gait. The handler may also use the elastic band to advance the limb and manually place the foot as the animal walks on the treadmill. A rigid toe extension attached to a boot, syringe cap, or any mild irritant (such as a marble or bottle cap) placed on the bottom of the contralateral, unaffected foot may encourage weight bearing in patients that are reluctant to place weight on a limb (Figure 29-11).

Aquatic Therapy

In addition to standing in water, patients that have some motor function may greatly benefit from walking in water or on an underwater treadmill. The buoyancy of the water, combined with ambulation, results in support while walking. Further support may be provided by using life vests, slings, harnesses, or an assistant in the water. Elastic bands may be used to help advance limbs in the water. After each aquatic use, the bands should be laid flat and allowed to thoroughly dry to prevent degradation and tearing of the elastic. Assisted swimming is also possible in patients, especially smaller patients recovering



Figure 29-11 A syringe cap, rigid toe extension attached to a boot, or any mild irritant (such as a marble or bottle cap) placed on the bottom of the contralateral, unaffected foot may encourage weight bearing in patients that are reluctant to place weight on a limb.

from neurologic conditions, such as intervertebral disk herniation surgery. Patients with little motor function may simply kick the limbs in an uncoordinated fashion in the beginning, but such activity results in muscle contraction and strengthening and promotes recovery of neuromuscular coordination and function. It is important to provide assistance and support and not allow vigorous swimming initially. As the patient improves, the assistance is gradually reduced.

Aids for Dynamic Ambulation

Walking Slings

Commercially available canine slings should be durable, flexible, and washable and should conform to the body. They are available for the forelimbs and hindlimbs and for trunk support. When using rear limb slings it is important that bowel and bladder function are not obstructed. Adjustable-length handles for the therapist to grip make a sling a more ergonomic choice of assistive device than a bath towel because the sling allows the handler to stand erect without bending forward while carrying the weight of the dog and reduces excessive stress to the handler's forearm, wrist, shoulders, and back. This is an important concern, especially if the handler will be assisting a large dog several times a day for any length of time. It is important for persons handling the animal to practice proper body mechanics while assisting the canine patient to avoid injuring themselves. The ideal sling length and position for the handler is to simulate carrying a suitcase or large bag, and not bending over to stress the back, while making sure the patient is able to position the paws in a weight-bearing position without dragging the toes or dorsum of the paw on the ground. The sling may be used for both orthopedic and neurologic patients in need of assisted ambulation.

Towels

A long towel may be used as a last resort to support an animal that requires assistance with standing or ambulation. Towels are readily available, inexpensive, and washable but can be problematic and are often more difficult than slings with handles for the therapist or owner, especially in patients requiring long-term assistance. When assisting the dog's hindlimbs, the handler should place the towel around the abdomen just in front of the hindlimbs in the inguinal region. Nylon straps are a highly recommended modification and may be sewn onto the ends of the towel to provide added length so the handler can maintain proper body mechanics.

A disadvantage of towels on larger dogs with urinary incontinence is that it puts pressure on the urinary bladder and may cause it to be expressed. This is less likely to happen with a properly fitted commercial pelvic sling.

Elastic Bands

It is often beneficial to “graduate” a patient from a more supportive sling into a more dynamic sling. Elastic bands used for gait patterning or resistance exercises are an excellent means to provide some support and concurrent dynamic feedback during weight bearing. There are several levels of resistance and thickness that should be matched to the patients' size and functional ability. Lighter weight bands are ideal for smaller patients even as an initial sling. Heavier weight slings can be used in medium to large dogs with moderate functional ability both as a support and a gait challenge. The elastic band in this stage can be used to pull the patient off balance to one side, while providing enough support to “catch” the patient so it doesn't fall during walking or standing exercises. Bands may also be attached to harnesses to provide resistance during gait.

Carts

Mobility carts are appropriate for temporary use as the animal regains its ability to ambulate independently, as well as for permanent use for animals that will not regain full function. After a period of acclimation for both the animal and owner, a properly fitted cart may result in a positive change in the quality of life for both. When the animal is mobile in a cart, that mobility fosters independence that may have a profound positive effect on the animal's attitude. Carts with fabric sling supports can be readily adjusted to allow preferential support or challenge to individual limbs. Additionally, elastic bands tied to the cart can be used to provide additional resistance for individual limbs or to help a weak limb advance, similar to elastic band use on land treadmill exercises.

The home environment may need to be altered to allow the animal to freely ambulate in the cart without encountering dangerous conditions. The animal's exercise area should be free of obstructions that could hinder the animal's ability to safely ambulate in the cart. Particular caution should be taken to prevent falls down stairs.

Independent Ambulation

Leash Walking

Slow leash walks are perhaps the most important exercise in the early rehabilitative period, and they are commonly performed incorrectly. Walking the animal slowly encourages the use of all limbs in a sequenced gait pattern. Walks must be slow enough to allow weight bearing; if the dog is walked too fast, the tendency is to simply hold the intended limb up in a flexed position and not bear any weight on it. Slow leash walking is indicated when the animal is reluctant to use a limb as a result of pain, weakness, or proprioceptive deficits. Slow leash walks encourage placement of each limb on the ground, increasing stance time and weight bearing.

If there are no contraindications to weight bearing, slow leash walks may begin very soon after most orthopedic procedures. Behavior modification is important. The dog should be praised when touching the limb to the ground, and not praised when the leg is held up. As the animal regains use of the affected limb and is consistently able to place the limb at a slow leash walk, the pace of the walk may be increased. Faster walks further challenge balance, coordination, proprioception, and cardiorespiratory endurance, as well as functional muscle strengthening and endurance. When an animal is weight bearing well at slow and fast speeds, an additional challenge is to focus on speed transitions while walking. Often the patient will have difficulty increasing or decreasing speeds, which requires significant muscular control and coordination in animals recovering from surgery, neurologic conditions, or significant musculoskeletal injuries. Animals recovering from neurologic conditions may be challenged by gently pushing the animal off-balance while walking, using the handler's leg or an elastic band sling, thus encouraging proprioceptive development. The therapist must be certain that the animal is not unbalanced so much that it falls, however. When appropriate, the therapist may alter the exercise treatment plan to include fast walking, slow jogging, and running on a long lead.

Treadmill Walking

Treadmills may be useful during rehabilitation programs as a continuation of assisted ambulation or for conditions in which extension of the hip or stifle is painful, such as hip dysplasia or postoperative recovery from cranial cruciate ligament surgery. Normally, patients are reluctant to perform activities such as climbing stairs, because extension of these joints is painful. Treadmill walking is less painful in some patients because the belt provides assistance with hip and stifle extension by helping to pull the rear limb caudally. There is less need for active contraction of the gluteal and quadriceps muscles for joint extension when walking on a treadmill than when walking on land. The treadmill may be angled up or down to reduce or increase the forces placed on the forelimbs or hindlimbs.



Figure 29-12 Walking or jogging the dog up inclines aids in strengthening of the quadriceps, semitendinosus, semimembranosus, and gluteal muscles with relatively low-impact activity.

The speed and length of treadmill walking may be increased as the animal recovers and gains strength and stamina. Also, varying intervals of high and low speeds within a session may be introduced.

Inclines and Declines

Walking or jogging the dog up inclines aids in strengthening of the quadriceps, semitendinosus, semimembranosus, and gluteal muscles with relatively low-impact activity (Figure 29-12). Rear limb muscle strength is required for the dog to propel itself up an incline, whereas forelimb muscle strength and eccentric contraction of both forelimb and hindlimb muscles are required for the dog to brake itself while going down a declined surface. Walking should be done slowly and on leash; otherwise the dog may only toe-touch with the limb or hop in a non-weight-bearing fashion. In addition, if the head is held up slightly while walking up an incline, the weight will be shifted caudally, requiring the animal to drive up the hill with the rear limbs and use the muscles to a greater extent. Conversely, if the head is held in a downward position, the weight will be shifted cranially, requiring the forelimb muscles to engage for braking. It is often easier to control the dog's head and body position on inclines if wearing a halter and harness rather than a neck lead.

Weight bearing while climbing promotes strengthening of the quadriceps and gluteal muscles. When the limb is in the stance phase of gait and the body is traveling forward, the stifle and hip must extend to propel the animal forward. If extension is painful, the dog's stance time on the limb may be shortened and an altered gait may result. Inclines and declines should be introduced slowly, beginning with gradual slopes. As the dog's range of motion and strength improve, the dog may be challenged by walking up longer, steeper slopes and by increasing the duration and speed of the climbing exercise.

Walking down declines is useful for strengthening the triceps muscles, and is typically more difficult for hindlimb

conditions because it requires the dog to reach under the body with the hindlimbs, requiring flexion of the hock, stifle, and hip. Start with gently sloping, short declines and progress to steeper ones as the dog is able. If an incline or decline is too challenging, an easier introduction is to walk on a gently sloped hill in a serpentine pattern or on the diagonal up, down, or both until the patient is strong enough to navigate the incline and decline in a straight line. Additionally, diagonal climbing and descent can be used to alter the challenge to an affected limb depending on if the limb is uphill or downhill. For example, if the affected limb is on the downhill side while ascending diagonally, the joints and muscles perform with the limb in a more extended position. Conversely, if the affected limb is on the uphill side while ascending diagonally, the joints and muscles perform with the limb in a more flexed position.

Other Methods to Encourage Weight Bearing

In some situations, a patient may be reluctant to use a limb after injury or surgery, even though the stage of recovery is such that limb use would normally be expected. It is important to always rule out any pathologic causes of poor limb use, such as failed implants, infection, other concurrent conditions, and so on. If there is no pathologic cause of poor limb use, and pain is adequately controlled, other measures may be used to encourage limb use.

A syringe cap, coin, or marble may be placed on the bottom of the uninjured foot and held in place with adhesive tape to encourage weight bearing on the affected limb (see Figure 29-11). Also, creating limb length differences with rigid or semistiff toe extensions attached to the foot with an elastic wrap or attached to a boot may also be used for gait perturbations. In many instances, weight bearing continues on the affected limb even after the item is removed from the good foot. Some dogs will build up a tolerance to the item; therefore continuous application is not recommended. In some particularly difficult cases, a nonweight-bearing sling may be placed on the nonaffected limb to force the animal to bear weight on the affected limb. In our experience, this technique is less effective because many animals simply lie down and will not attempt to walk.

Side bending and vertical head movements to follow a toy or treat are often useful to encourage affected limb use. The patient should be positioned in an even, square stance with all four limbs and slowly asked to bend to the left and right, with the goal being to touch the nose to the stifle. For vertical movements, the dog should reach up slowly to a comfortable point and then down, with the goal being to position the head between the forelimbs. This exercise not only enhances spinal flexibility but also redistributes body weight to encourage weight bearing and position awareness of the affected limbs. Caution should be exercised in patients with nonsurgical spinal injuries or postoperative

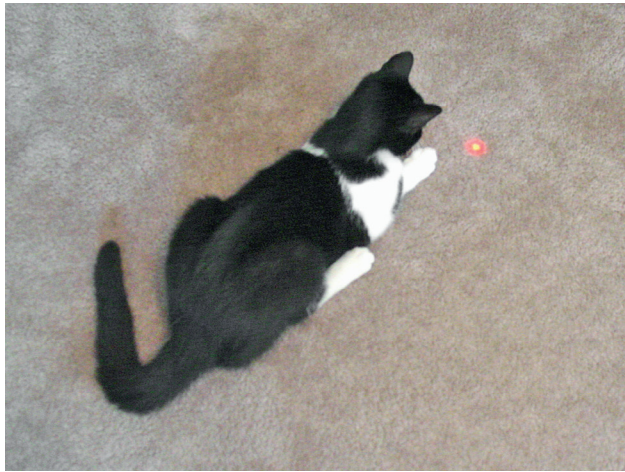


Figure 29-13 Many cats will readily chase the spot of light from a flashlight or laser pointer around a room. Care should be taken to have good footing, such as carpeting.

spinal cases, and movements should be minimal and well within the voluntary range of comfort for the patient. This technique is contraindicated in animals with an unstable spinal canal or those with unstable intervertebral disks.

Hemiwalking is sometimes useful to encourage weight bearing on a particular limb. This exercise is performed by elevating the ipsilateral fore- and rear limbs off the ground and shifting the weight to the contralateral limbs. The animal must move the contralateral fore- and rear limbs laterally to avoid losing balance and falling. Although this is generally more successful than placing a sling on a limb, some animals may still resist and simply lie down.

Many cats will readily chase the spot of light from a flashlight or laser pointer around a room. Care should be taken to have good footing, such as carpeting, and to move the light at a speed appropriate for the animal's stage of recovery (Figure 29-13). Feathers and reflective material on a string or stick are also useful as a lure in feline rehabilitation. In addition, stretching may be employed by moving the light or feather along a vertical surface, such as a wall. Some dogs will also chase a laser pointer light. Caution is advised using this technique because some animals, particularly dogs, may become obsessive-compulsive and develop chronic or permanent behavior problems when encouraging light-chasing exercises.

Many cats and dogs have been well conditioned to the sound of a can opener or other audible cues such as a door opening to go for a walk. The animal may be placed in another room and the sound made. In many instances, the animal disregards a lame leg and ambulates to the sound.

Most cats and some dogs will play with a toy such as a string or other object that they are used to "stalking" or chasing. Engaging the owner to identify some favorite playthings is important, because some animals will not react to toys in a strange environment such as a clinic. It is equally important to instruct the owner regarding

the length of playtime, how to use the toy to maximize the rehabilitation process, and limits of play to reduce the chance of fatigue and injury.

Progression of Exercises

Many variables must be considered in the design of a therapeutic exercise program, including the type and severity of the condition; the stability of any surgical repairs; the number of involved limbs and joints; the size of the animal; the preexisting physical condition of the patient; the available facilities; the skill and experience of the therapist; and whether the exercises will be administered by a professional, the owner, or a combination of the owner and a professional. Unfortunately, it is impossible to have a single protocol for each specific condition. Some basic guidelines may be applied, however.

The goal of any therapeutic exercise program should be to restore the animal to as full and active use as possible. The goal may need to be periodically adjusted if the patient progresses to a greater or lesser degree than expected. The initial goal should be to have the patient bear its full weight while standing. The next step should be assisted active ambulation. Proprioceptive training should occur concurrently with active ambulation, with the goal to have unassisted active ambulation while maintaining balance. As the animal nears return to house-pet function, strategies should be developed to encourage increased weight bearing and muscle mass.

It is very important to perform the exercises correctly and to maintain a consistent level of activity daily or every other day. In most cases, several short sessions per day will be more effective than a single long session early in the recovery period. Overdoing activity on the weekends with relatively little activity during the rest of the week will likely be detrimental to the animal's recovery; regular, daily activity will avoid the "weekend warrior" syndrome.

A general rule of thumb is to increase the activity by 10-15% each week. If pain or lameness increases during or several hours after stepping up an activity, the level of activity should be reduced by 50% for 3 to 7 days. If the lameness and pain resolve, the activity may be slowly increased back to the pre-step-up activity level over 3 to 5 days, with careful observation to be certain that the activity is not increased too aggressively or rapidly. The rate of subsequent step-up activity should be half of what it originally was to avoid a recurrent episode of exacerbated lameness and pain.

It is also important to vary the routine for the animal's and the owner's sake to avoid boredom, and to allow incorporation of function-specific activities in the animal's home or work environment. During chronic management of the patient, weight control is extremely important. It is also important to continue to oversee the protocol and document improvement. Progress will occur at a slower

rate during the chronic phase of recovery, but continued rehabilitation is just as important to achieve as full a return to activity as possible.

Several medications may be appropriate for use during rehabilitation, including nonsteroidal antiinflammatory agents and chondroprotective agents. Caution should be used, however, to be certain that any analgesic medications do not mask pain and discomfort associated with the level of activity. They should facilitate rehabilitation and recovery rather than allow rehabilitation to progress faster than normal conditions and tissue healing should allow. When stepping up the level of an activity, antiinflammatory or pain medication may be withheld on the day of step-up and for the day after, to be certain that there is no additional tissue damage. If pain or lameness increases, the level of activity should be reduced 50% for 1 week; then the activity is stepped up again at a slower rate.

Summary

Beginning a rehabilitation program need not be elaborate or costly. Consideration should be given to the patient's needs, the owner's needs, and the therapist's needs. Also, although protocol development greatly depends on the available facilities and equipment, the willingness of the owners to help with rehabilitation, and the education level of the staff, some rehabilitation is better than none at all.

Therapeutic exercises are undoubtedly the most important aspects of rehabilitation. They may (and should) be incorporated with other modalities to enhance recovery, but it is the exercises that help improve muscle strength, joint mobility, limb use, proprioceptive training, and endurance to maximize function. Although a variety of techniques have been described, the ingenuity of the rehabilitation team, including the owner, will allow the development of other exercises that are specific to a patient's recovery. The keys to a successful therapeutic exercise program are to have site- and condition-specific exercises whenever possible, to use a variety of exercises and techniques to keep the therapy team and patient from becoming bored, and to allow the animal to appropriately progress so that tissues are adequately challenged for strengthening, but not so rapidly as to result in complications and tissue damage.

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30 Therapeutic Exercises: Joint Motion, Strengthening, Endurance, and Speed Exercises

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Therapeutic exercises to improve active joint motion through a more complete range, and to build power, strength, and speed are the final phases of rehabilitation to return the patient to as normal a function as possible. The therapist must keep in mind the degree of injury or disability in determining the likelihood for the patient to reach higher levels of function and adjust the rehabilitation program and expectations accordingly. For example, it is not expected that a 7-year-old dog with severe bilateral elbow arthritis would be able to achieve near-normal function, but there are exercises that can be performed to help achieve greater function and ability. Some of the common goals of therapeutic exercise are to improve active pain-free range of motion (ROM), muscle mass and muscle strength, performance, and conditioning. Common activities include stair climbing, sit-to-stand exercises, treadmill activity with resistance, pulling or carrying weights, “wheelbarrowing” (for forelimb activity), “dancing” (for rear limb activity), walking and trotting across cavaletti rails, jogging, running, underwater treadmill activities, swimming, playing ball, and activities using balance balls or rolls. These higher levels of therapeutic exercise are important in assisting an animal’s return to the best function possible following early limb use and proprioceptive exercises. Many of the exercises that target certain aspects of performance, such as joint motion, conditioning, and strength, have some overlap (Box 30-1). In addition, the equipment needed for most exercises is relatively inexpensive, and similar principles apply to a variety of individuals and conditions. Therapeutic exercise programs designed for the home environment also provide an opportunity for owners to become actively involved in their pets’ rehabilitation. In any case, it is very important to have good footing to avoid slipping and to have adequate space to perform exercises with as little distraction as possible.

It is also important to remember to use patience when working with the dog to learn new exercises. Most of the exercises in this chapter have been purposefully selected so that most dogs that are trained to leash can perform them with some basic training. However, some dogs are highly trained and the therapist should work with the owner to select exercises based on the patient’s needs, as well things the dog is already trained to perform. The therapist is cautioned to use training techniques that are based on

positive reinforcement. Negative reinforcement techniques may temporarily work in some instances, but because rehabilitation is an ongoing treatment, dogs will become less willing to perform an activity or to learn new ones. Many dogs will be somewhat nervous or frightened when first undergoing rehabilitation, especially if pain is involved from their injury or surgery. It is important to go slowly and gently at the beginning. Treats, favorite toys, or other rewards should be kept handy to encourage dogs when learning new exercises. The reader is referred to Chapter 4 for further tips on applying behavior techniques in the rehabilitation environment.

When designing a high-level therapeutic exercise program, several factors must be considered. A problem list is developed based on an initial evaluation of the rehabilitation patient, and a treatment plan is formulated to address the identified problems. Realistic outcome goals are then established. Appropriate exercises are those that can be performed safely and effectively and accomplish the therapeutic goals. The exercise plan should target the affected muscles, joints, and other structures. When prescribing therapeutic exercise, the therapist should understand the diagnosis, identify the structure or structures involved, and recognize the stage of tissue recovery with the resultant functional limitations. Without a good understanding of these principles, it is easy to over stress the involved tissues and potentially cause harm. The type of injury or repair must be considered when formulating a treatment plan. For example, a patient with a tibial plateau leveling osteotomy (TPLO) will require several weeks for tissue remodeling and healing of the osteotomy before engaging in high-speed activities. Similarly, a patient with a partial brachial plexus injury may initially have a difficult time performing a shaking hands exercise. But over time, the dog may recover and perform shaking hands exercise with a limb weight to achieve greater strengthening. With this knowledge and understanding, appropriate decisions can be made regarding therapeutic exercise choices.

Treatment considerations and choice of exercises vary with each stage of tissue repair and endurance. As the animal improves clinically and tissue healing progresses, the exercise plan should be altered to match the animal’s progress and appropriately challenge the involved tissues. The intensity of an exercise may be increased or reduced

Box 30-1 Exercises to Improve Joint Motion, Conditioning, and Strength**Joint Motion Exercises**

Incline and decline walking
 Stair climbing
 Dancing
 Wheelbarrowing
 Sit-to-stand exercises
 Cavaletti rails
 Underwater treadmill activities
 Swimming
 Tunnel and limbo walking

Conditioning Exercises

Incline and decline walking
 Stair climbing
 Treadmill activity
 Jogging
 Running
 Underwater treadmill activities
 Swimming
 Playing ball
 Sport-specific activities

Strength Exercises

Incline and decline walking
 Stair climbing
 Treadmill activity with resistance
 Dancing with resistance
 Wheelbarrowing with resistance

Running up stairs or inclines or down declines
 Underwater treadmill activities with speed or resistance, floatation devices
 Swimming
 Pulling or carrying weights
 Playing ball
 Activities using balance balls or rolls (core strengthening)
 Rhythmic stabilization
 Lateral hopping with resistance
 Sit-to-beg for core strengthening
 Sit-to-stand with backpack or weights
 Jumping
 Limbo walking
 Sport-specific activities

Feline Exercises

Proprioceptive exercises (e.g., weight-shifting activities, balance board, wobble board, and standing on a therapy ball or roll)
 Walking with a harness
 Ground treadmill walking with support and assistance
 Treadmill walking with small leg weights or therapy bands
 Playing with feathers or toys tied on the end of the string
 Scratching post
 Can opener
 Low cavaletti rails
 Chasing a flashlight beam or laser light (use caution to avoid obsessive-compulsive light chasing)
 Aquatic therapy

by changing the duration of time that an animal performs an exercise, the rate of speed at which a particular exercise is performed, or the frequency (how many times per day) that it is performed.

These factors may be altered to fine-tune an exercise prescription to achieve the expected outcome goals. For example, a realistic initial goal for a morbidly obese, deconditioned patient with degenerative joint disease may be to increase the amount of time the dog is able to comfortably tolerate walking, thus improving endurance and promoting weight reduction. Increasing the speed at which the dog walks may not be a realistic initial goal for this animal. A contrasting example may be an athletic animal recovering from injury that must be challenged to improve speed and frequency to meet the goal to return to a high level of performance. It is important for the therapist to have an understanding of exercise intensity and what is appropriate for each patient during rehabilitation treatment. Routine reevaluation of the patient is recommended to assess the adaptations that are occurring with the rehabilitation treatment plan and to determine the appropriate rate of progression.

Therapeutic exercise routines should be monitored at regular intervals by a trained individual familiar with the patient and the exercise techniques. Inappropriate exercise

or improper technique may result in inappropriate stresses, further injury, or exacerbation of an existing condition. Certain exercises may not be safe for the strength, flexibility, or endurance level of the animal performing the exercise, or an exercise may not be the correct one to accomplish the intended goal. The exercises described in this chapter are listed with the desired outcomes of joint motion, conditioning, or speed and strength, but there is significant overlap with some exercises, and the differences are indicated for the various outcomes. The reader is also referred to Chapter 24 on biomechanics of rehabilitation for further details on therapeutic exercises, especially as they relate to joint motion and forces.

Joint Motion Exercises**Incline Walking**

Walking the dog up an inclined surface alters joint motion and aids in strengthening of the hip and stifle muscles, especially the extensors, while ascending (Figure 30-1).

Joint Motion Considerations

Climbing up an inclined surface promotes increased flexion, extension, and ROM of the hip. Stifle flexion is

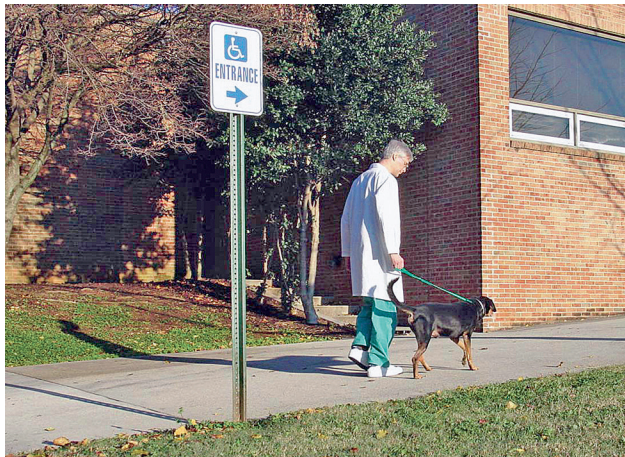


Figure 30-1 Leash walking up ramps or hills encourages muscle strengthening and cardiovascular endurance.

decreased, however.¹ When the limb is in the stance phase of gait and the body is traveling forward, the knee and hip must extend to propel the animal forward. If extension is painful, the dog's stance time on the limb may be reduced and an altered gait may result. In addition, dogs may shorten the stride length to help control joint motion if a particular joint position causes pain. For example, a dog with hip dysplasia may shorten its stride while walking up an incline so that excessive hip extension and pain are avoided. If this occurs and a particular joint motion that is desired does not occur, it may be better to use another exercise, such as stair climbing, in which the limb must travel a discrete distance to negotiate the step rather than adjust the stride length while walking up a ramp.

Walking up an inclined ramp also results in greater ROM of the forelimb joints.² Shoulder extension and ROM, elbow flexion and extension, and carpal flexion are greater when walking up a ramp as compared with ascending stairs or trotting over level ground.

Exercise Considerations

Initially, walking should be done slowly and on leash; otherwise the dog may only toe-touch with the limb or hop in a non-weight-bearing fashion. Inclines should be introduced slowly, beginning with gradual slopes. As the dog's ROM, strength, and limb use improve, the dog may be challenged by walking up longer, steeper slopes and by increasing the duration, speed, and frequency of the climbing exercise. Further challenges are provided by doing inclines while wearing a weighted vest, leg weights (Figure 30-2), or pulling a small amount of weight with a harness. Ascending or descending a slope at an angle may also alter joint motion of an affected limb. If the affected limb is on the downhill side while ascending diagonally, the joints and muscles perform with the limb in a more extended position. Conversely, if the affected limb is on the uphill side while ascending diagonally, the joints and muscles



Figure 30-2 Walking up a ramp or hill while wearing a leg weight provides greater challenges.

perform with the limb in a more flexed position. It is particularly important to monitor gait patterns when walking downhill as many dogs with neurologic conditions may exhibit more ataxia, causing the hind end to sway to one side or the other. Also, if descending too fast, the patient may stumble and fall even if its gait appears to be normal on level ground. Varying speeds up and down inclines is another method to increase the challenge of this exercise.

Stair Climbing

Climbing stairs is useful to improve ROM, power in the rear limb extensors, coordination, and proprioception, and in many instances, is an activity of daily living that must be performed in the home environment (Figure 30-3). Quadriceps and gluteal muscle groups are strengthened as the animal pushes off, extending the stifles and hips while propelling the body up the steps.

Joint Motion Considerations

ROM of the pelvic limb joints is significantly increased during stair climbing compared with trotting on level ground.^{3,4} Extension of the hip and hock joints is moderately increased with stair climbing, whereas stifle extension may be mildly increased or decreased, depending on the size of the dog relative to the size of the step. Flexion of the stifle and hock is also increased. Although hip flexion contributes little to stair ascent, the stifle and hock joints seem to contribute significantly to ascending stairs by flexing these joints to raise the limb up to the height of the step.

Regarding the forelimb joints, there is less ROM with stair ascent as compared with walking up a ramp with similar slope.² These differences are primarily due to changes in extension, with no significant differences in flexion of forelimb joints. Shoulder extension is less with stair climbing as compared with trotting over ground,



Figure 30-3 Stair climbing should begin on stairs with a gradual rise (A) and progress to steeper stairs (B). Care should be taken to be certain that the dog pushes symmetrically off of both rear limbs.

whereas flexion is greater with stair climbing. Elbow and carpal flexion, extension, and ROM are greater with stair ascent as compared with trotting over ground.

Stair descent results in significantly greater stifle flexion and hock flexion and extension compared with walking down a declined surface.⁵ Significantly greater ROM also occurs in the coxofemoral, femorotibial, and tibiotarsal joints during stair descent.

Exercise Considerations

The dog must begin slowly climbing stairs to encourage proper use of the rear limbs, as opposed to simply carrying the limb, hopping with both rear limbs, or skipping up stairs. Encourage the dog to go slowly and deliberately, climbing the stairs in a reciprocal stepping gait. Stairs should be introduced slowly because this is a challenging exercise for both the musculoskeletal and cardiovascular systems and the animal may fatigue quickly. Initially, some dogs may require assistance from the handler. Begin with five to seven steps, and gradually increase to two to four flights of stairs once or twice daily. In severely affected dogs or smaller dogs, it is advisable to begin stair climbing on stairs with a gradual ascent and progress to steeper stairs. Another strategy for smaller dogs is to attach blocks to existing stairs to make “half stairs.” As the patient improves, dogs may jog up stairs. In the motivated patient, walking backward up or down stairs can be an additional significant challenge. To introduce this exercise, use a treat or toy to encourage stepping backward up one step while the patient keeps the forelimb on the ground and then have patient come back down to level ground. The reverse is true for backing down a step. Repeat this several times until the patient is comfortable with this movement before having the patient walk backward up or down two to three steps, and advance to a full flight after the patient is comfortable and understands what is being asked. This

advanced exercise is often easier to teach using a few steps versus a full flight of stairs. A three- or four-stair block can be built for this purpose.

Dancing and Wheelbarrowing

Raising the forelimbs off the ground and walking the dog forward or backward, also called *dancing*, alters motion of rear limb joints and challenges proprioception, coordination, and strength. The amount of peak force placed on the limbs is slightly greater than what would occur at a trot.⁶ Therefore if the stage of recovery is not adequate to withstand trotting, then dancing should not be undertaken. On the other hand, additional strengthening of postural muscles may be obtained with dancing when it is safe to perform the exercise.

Wheelbarrowing is an exercise similar to dancing, except that the forelimbs are targeted by raising the rear limbs off the ground and walking the dog forward. This exercise encourages increased use of the forelimbs and challenges proprioception, coordination, and strength. The dog’s orthopedic condition must be adequately stable to handle the stresses of these exercises. Wheelbarrowing results in peak force placed on the limb that is between walking and trotting.⁶

Joint Motion Considerations

Forward dancing results in significantly less hip flexion, total hip ROM, hock flexion, and hock total ROM compared with walking.⁶ Dancing backward results in greater hip extension and stifle flexion, whereas hip flexion, hip ROM, stifle extension, and stifle ROM are less than regular walking.

Wheelbarrowing alters motion of the forelimb joints, increasing shoulder extension and elbow flexion as compared with walking.⁶



Figure 30-4 Dancing encourages weight bearing on the rear limbs. Muzzling may be necessary for some dogs before this exercise, particularly in dogs that do not like to have their feet touched or feel threatened by the therapist's posture. The forelimbs are lifted off the ground, allowing the patient to bear weight only on the rear limbs.

Exercise Considerations

It is important for the therapist or veterinarian to evaluate available ROM in the rear limb joints before attempting dancing to identify any potential limitations that may prevent the animal from safely performing the exercise. For example, in the early phases of a therapeutic exercise program for a dog that is painful with hip extension as a result of hip dysplasia, gluteal muscle strengthening may be more comfortable by dancing the dog forward rather than backward.

When a dog is using its affected limb consistently at a walk with minimal lameness, dancing may begin. Muzzling may be necessary for some dogs before this exercise, particularly in dogs that do not like to have their feet touched or feel threatened by the therapist's posture. The forelimbs are lifted off the ground, allowing the patient to bear weight only on the rear limbs (Figure 30-4). Dogs with normal proprioception will naturally move the rear limbs as the handler moves and the animal "dances" backward and forward. Some dogs may resist dancing forward if the handler stands in front of the dog; dogs may plant their rear limbs and stretch out as the handler moves until the forelimbs reach the ground. In this situation, the handler should stand behind the dog, placing the arms under the axillary region to support the dog, and walk forward. How far the dog is elevated off the ground depends on the amount of stress the animal is able to comfortably handle on the rear limbs. Dogs may be elevated as high as possible and also dance up and down inclines or hills to place additional stress on the rear limbs. Dancing activities may



Figure 30-5 Dancing activities may also be accomplished with assistance of a physioroll or egg ball. The patient places their front limbs on the roll or ball and the therapist slowly rolls the ball toward or away from the patient forcing them to move their hindlimbs backward or forward.

also be accomplished with the assistance of a physioroll or ball. The front limbs of the patient are placed on the roll or ball and the therapist slowly rolls the ball toward the patient, forcing the dog to move the hindlimbs backward (Figure 30-5). Most dogs will reposition their forelimbs, especially if they are standing on the ball, rather than resting their antebrachium on the ball. Once comfortable with walking backward in this fashion, the ball can be slowly rolled away from the patient, stimulating forward walking of the rear limbs. A harness to provide support for the patient is very helpful for this activity. Alternatively, a wagon or other flat surface on wheels can be used for this activity, particularly for forward dancing.

To perform the wheelbarrow exercise, the handler places the hands under the caudal abdomen and lifts the rear limbs of the dog off the ground, and the dog is moved forward (Figure 30-6). The handler may wish to place a muzzle on the dog. Dogs with normal proprioception will move the forelimbs so they do not fall. Some dogs may require sling support if they are weak. Dogs may be wheelbarrowed up and down inclines for greater muscle strengthening (Figure 30-7). Theoretically, the higher the animal is lifted, the greater the forces placed on the forelimbs. However, the increase in weight bearing may not be as great as anticipated because the stride length is much shorter when the animal wheelbarrows from a greater height. The shortened stride results in less force placed on the limbs while wheelbarrowing as compared with walking or trotting at a faster speed. A hindlimb sling, particularly one that lifts at the pubis rather than the abdomen, may be used to assist the dog while wheelbarrowing.



Figure 30-6 Wheelbarrowing encourages use of the forelimbs. The therapist supports the abdomen and lifts the rear limbs of the dog off the ground, and the dog is moved forward. The handler may wish to place a muzzle on the dog.



Figure 30-7 Dogs may be wheelbarrowed up and down inclines for greater muscle strengthening.

Sit-to-Stand Exercises

Sit-to-stand exercises help strengthen hip and stifle extensor muscles and improve active ROM. The act of sitting, then standing up, requires muscle strength of the gluteal, quadriceps, hamstring, and gastrocnemius muscle groups.

Joint Motion Considerations

Sit-to-stand exercise results in greater flexion of the tarsus and nearly twice as much flexion of the stifle and hip, whereas extension of these joints is less as compared with walking.^{7,8}

Exercise Considerations

Some training may be necessary, and low-calorie treats may be offered as a training aid to provide motivation to perform the movement. It is important to perform these exercises correctly. Attention should be paid to sitting and standing straight, with no leaning to one side, and the joints of both rear limbs should be symmetrically flexed so that the dog sits squarely on its haunches (Figure 30-8). While on the leash, after a sufficient warm-up period of walking, the handler asks the dog to sit squarely for a few seconds and then asks the dog to stand, take a few steps forward, and then again sit. The sit-to-stands may be repeated a number of times before the dog is allowed to rest. It may be easier in some cases to back the dog into a corner, with the affected limb next to a wall so that the dog cannot slide the limb out while rising or sitting. Start with 5 to 10 repetitions once or twice daily, and work up to 15 repetitions three to four times daily.

This exercise may be particularly beneficial for dogs with osteoarthritis of the hips. These patients generally feel pain when the hip joints are extended. Going from a sitting position to a standing position results in movement of the hip joint from a greatly flexed position to a hip angle similar to normal standing. In addition, there may be atrophy of the gluteal muscles. The sit-to-stand exercise allows active contraction of the gluteal muscles, but the hip joint is not generally extended to the point that results in pain. Therefore strengthening of the gluteal muscles can occur without the excessive pain encountered with hip extension.

Of course, each patient must be assessed to be certain that sit-to-stands are not too stressful or painful for that individual. If normal sit-to-stands are too painful, or the patient is too weak to rise from a sitting position, several modifications may be tried. Doing sit-to-stands with the pelvis in an elevated position may be helpful in patients that are too weak to rise from a complete sitting position. Dogs may sit with their pelvis on a stool, step, stack of books, curb, firm cushion, or the therapist's leg or knee, and rise from this elevated position. Many dogs will need guidance from the therapist's hands to learn to sit on an object because it may feel strange for some patients. Alternatively, dogs may be assisted with a sling from a sitting to a standing position. Dogs may also do sit-to-stands on an incline, with the body facing downhill to make it easier to stand. Alternatively, dogs may do sit-to-stands on an incline with the body facing uphill or sideways to each direction to provide an additional challenge and greater strengthening. For this challenge, dogs may perform five repetitions in each direction, twice daily. Finally, dogs may find it easier to do sit-to-stands in the water, using the buoyancy of the water to help assist them to a standing position.

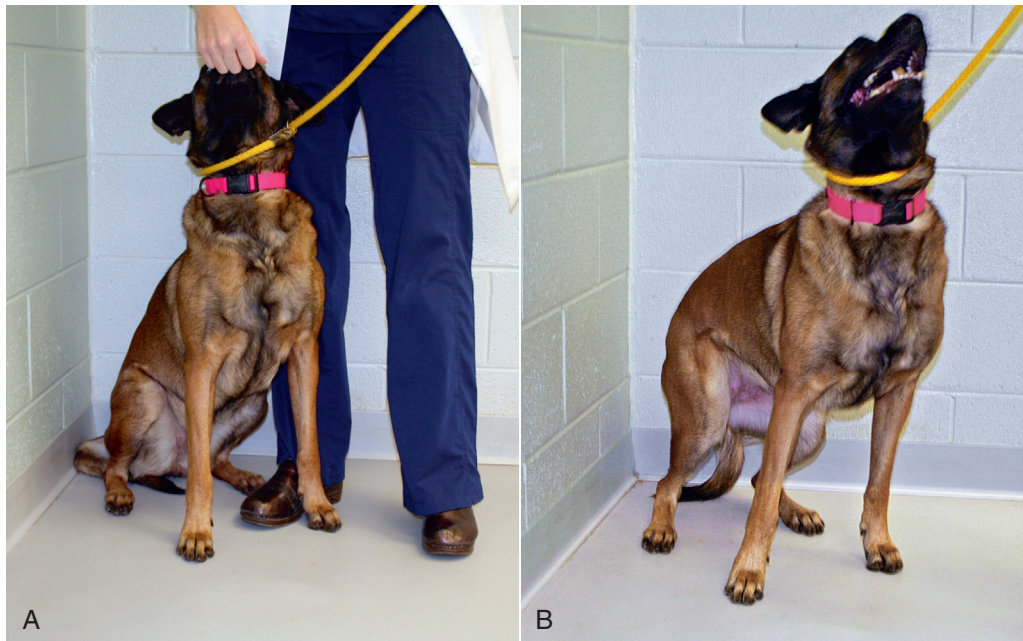


Figure 30-8 Sit-to-stand exercises. **A**, Note the symmetrical position of the dog in the sitting position. Placing the affected limb toward a wall will help the dog to sit in a square, symmetrical position. **B**, As the dog rises, it should push off equally with both rear limbs. Attention should be paid to sitting and standing straight, with no leaning to one side, and the joints of both rear limbs should be symmetrically flexed so that the dog sits squarely on its haunches.

Down-to-Stand Exercises

A variant of the sit-to-stand exercise is the down-to-stand. With this exercise, the dog is allowed to rise from a ventrally recumbent position to a standing position. In addition to strengthening the muscles of the pelvic limb, muscles of the forelimb are also engaged.

Joint Motion Considerations

In particular, the shoulder and elbow extensor muscles are activated through a greater ROM. The amount of motion has not yet been quantified, but joint motion of the hindlimbs should be similar to sit-to-stand exercises.

Exercise Considerations

It is important to have the dog rise symmetrically and push off equally with all four limbs. This exercise may require some additional time and training to perform correctly. Dogs will most commonly rise to a sitting position first and then stand, which is not the goal of this exercise. Similar to the sit-to-stand exercises, some training may be necessary with low-calorie treats to provide motivation to perform the movement. Attention should be paid to try to get the animal to use all four limbs to push up squarely and symmetrically. Start with 5 repetitions once or twice daily, and work up to 10-15 repetitions three to four times daily.

Cavaletti Rails

Cavaletti rails are poles that are spaced apart on the ground at a low height. Cavaletti rails may be used to encourage

greater active ROM of certain joints, and lengthened strides in all limbs. They may also be used to challenge proprioception, balance, and coordination in animals returning to function following neurologic impairment.

Joint Motion Considerations

Walking over cavaletti rails results in significant increases in flexion and total ROM of the elbow, stifle, and hock as compared with walking, and the increase is related to the height of the cavaletti rails.⁹ There is also a mild increase in hip flexion. There are no differences in joint extension, however.⁹

Exercise Considerations

Commercial cavaletti rails are generally used, although they can be constructed relatively easily. An alternative to cavaletti rails is to use a ladder and allow the rungs of the ladder to act as the low rails. Although ladders are readily available in most households, they have limited flexibility to change the distance between the rungs and the height that the animal steps over the rungs. This exercise can be beneficial for either orthopedic or neurologic patients in need of greater active flexion and ROM of targeted joints, improved voluntary motor control, and accuracy in placement of the limbs.

One or more poles may be used and should be spaced at appropriate distances apart, determined by the dog's natural stride length (Figure 30-9). The distance between the rails should allow for a single step; the exercise may not be as beneficial if dogs are allowed to take two or three

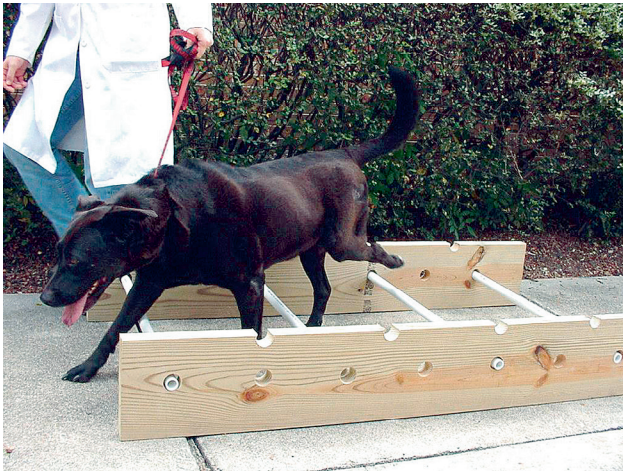


Figure 30-9 Cavaletti rails help increase active flexion of certain joints as the dog negotiates the rails. One or more poles may be used and should be spaced at appropriate distances apart, determined by the dog's natural stride length. Stride length may also be altered by placing the rails closer together or farther apart.

steps between rails. In general, the height of the poles should be at the same level or higher than the carpus initially, but severely affected dogs may need very low poles to start. For very small dogs, cavaletti poles, broom handles, or PVC pipe can be taped to the floor at an appropriate width for the patient. After the animal becomes accustomed to the task, the handler can further challenge the dog by making simple modifications such as adding more poles, increasing the height of all the poles to encourage greater active flexion and ROM of joints, and altering the heights of alternating poles to encourage dogs to negotiate different situations. Begin with walking 3-5 times over the rails, and progress to 8-10 times over the rails. To provide additional flexibility to the trunk and proprioceptive awareness, dogs may do serpentine walking in combination with cavaletti rail walking (Figure 30-10). Dogs may also progress to trotting for greater rate of joint motion and proprioceptive challenges, or the cavalettis can be positioned over an unstable surface such as dense foam.

Walking in Tall Grass, Sand, or Snow

Walking a dog through a field of tall grass enhances muscle strengthening and endurance because of the resistance provided by the grass, as well as coordination to navigate varying terrain. In addition, dogs have a tendency to flex their joints to a greater extent as they negotiate the grass, resulting in greater overall ROM.

Exercise Considerations

Some caution should be used in dogs with allergies walking in grass, which may be aggravated in such conditions, and in dogs with conjunctivitis, which might also be exacerbated. Walking in deep sand or snow provides resistance to



Figure 30-10 To provide additional flexibility to the trunk and proprioceptive awareness, dogs may do serpentine walking in combination with cavaletti rail walking.

limb movement during walking and jogging and strengthens the flexor muscles, which must contract more to lift the limbs. However, one must be cautious with walking in deep snow that is dense or crusty as dogs may have difficulty pulling their limbs from the heavy, wet snow. Also, some dogs with metal implants (for example, a TPLO plate) may experience discomfort or postexercise lameness after walking in snow because the metal may contract, resulting in discomfort. Similarly, walking and jogging in tall grass strengthens the extensor muscles to some degree, which must contract to a greater extent to advance the limb. Exercising in sand and snow minimizes concussive forces placed on arthritic joints, while allowing strengthening of supporting periarticular muscles. Caution should be used when starting these exercises to avoid muscle soreness. Short, frequent sessions are recommended in the beginning, with a gradual increase in the time and repetitions. Standing in sand may also be beneficial to dogs in rehabilitation following neurologic conditions.

Tunnels and Limbo

Agility training tunnels or children's play tunnels may be used to promote flexion in the forelimbs and rear limbs, while strengthening the supporting muscles that hold the joints in this more flexed position. The size of the tunnel opening should be appropriate for the size of the dog, that is, just shorter than the dog's standing height. The dog is encouraged to crouch low and walk through the tunnel, which requires greater limb flexion and strength than that



Figure 30-11 A telescoping pole, such as a shower rail, may be adjusted in a doorway, with the height lower than the dorsal midline. The dog then walks under the pole to perform the limbo exercise.

needed for normal walking. Greater challenges may be instituted by using tunnels that are even smaller as dogs strengthen their muscles.

Alternatively, a telescoping pole, such as a shower rail, may be adjusted in a doorway, with the height lower than the dorsal midline. The dog then walks under the pole to perform the limbo exercise (Figure 30-11). The pole may be lowered to challenge the dog to crouch to a greater degree as strength and ability allow.

Under and Over Exercises

A combination of walking over cavaletti rails and alternating limbo walking is termed *under and over exercise*. This is a more advanced exercise that challenges strength, joint flexion, and proprioception.

Circling, Figure-Eight Walking, Vertical Weave Poles

Circling aids in lateral flexion of the spine, strengthening of the adductor and abductor muscles, balance, and proprioception. Walking in figure eights is a progression of circling. It adds additional challenges by changing directions repeatedly and helps to prepare the patient for irregular turning as the patient returns to more normal function. Weaving between vertical poles also helps to promote side bending of the dog's trunk and also challenges proprioceptive functioning and strengthening of limb abductor and adductor muscles, but with more sudden changes in direction.

Exercise Considerations

The size of the circles for circling should be large to begin with, and as limb use and proprioception improve, the size



Figure 30-12 For vertical pole weaving, the distance between poles should be adjusted so that sufficient side bending results. The distance between poles should be slightly less than the body length of the dog. In addition, the handler must lead the animal so that the head, neck, and body actually flex as the poles are negotiated.

of the circles may become smaller. The same principles apply for figure-eight walking.

For vertical pole weaving, the distance between poles should be adjusted so that sufficient side bending results; in general, the distance between poles should be slightly less than the body length of the dog. In addition, the handler must lead the animal so that the head, neck, and body actually flex as the poles are negotiated (Figure 30-12). Alternatively, a very advanced version is to have the patient weave between the handler's legs, usually following a treat or toy (such as a Frisbee). The handler can then walk forward and backward while the patient continues to weave through the handler's legs. The patient should be released from exercise restrictions prior to performing this advanced exercise.

Aquatic Exercises

The reader is referred to Chapter 31 on aquatic therapy for specific information regarding aquatic exercise, especially underwater treadmill walking and trotting and swimming. In general, joint flexion is increased over typical land-based exercises, such as walking or trotting. With swimming, joint extension is less as compared with walking or trotting. Underwater treadmill walking has the advantage of increased joint flexion, with joint extension that is similar to land-based exercises. In addition, the amount of flexion of specific joints may be affected, to some degree, by the level of the water.¹⁰

Strengthening and Endurance Exercises

Many of the exercises used for strengthening and endurance are also used for joint motion exercises. The reader